

Andreas Lekas

Senior .NET Full-Stack Developer

andreaslekas9999@gmail.com (913) 732-7886 FL, United States

Profile

A **senior software architect and full-stack developer** with 13+ years of experience in designing, developing, and optimizing scalable and high-performance software solutions using **Microsoft .NET technologies**. Expertise in enterprise application architecture, **SOLID principles**, and **modern design patterns**, with a strong focus on scalability, performance, and maintainability. Proficient in **cloud-native development (Azure, AWS)**, frontend development (**Angular/React**), Agile methodologies, and test-driven development (TDD), with experience in CI/CD automation and DevOps practices to streamline development and deployment within **.NET environments**.

Professional Experience

Boston Consulting Group,

Senior .NET Full-Stack Developer

03/2020 – present

Boston, MA

- Collaborated with Tenet Health to develop a secure healthcare management system, using **.NET**, **C#**, and **Angular/TypeScript** to securely handle sensitive patient data. Implemented advanced encryption for full **HIPAA** compliance, reducing response times and improving operational efficiency.
- Engineered scalable **microservices** for a healthcare platform with **ASP.NET Core 6**, enabling flexible, independent service scaling to meet growing demand, reducing downtime, and boosting system performance.
- Enabled real-time telemedicine features via **Twilio** and **WebRTC**, creating secure, **scalable APIs** with **ASP.NET Core 6** to streamline patient-provider interactions, resulting in increased telemedicine adoption across the healthcare network.
- Designed and deployed event-driven architecture using **Azure Service Bus** and **Apache Kafka**, enabling real-time data processing and communication across distributed healthcare systems.
- Optimized database performance using **PostgreSQL** and **Entity Framework Core**, improving query efficiency through complex **query optimization** and implementing **Azure Cache for Redis** to reduce database load for frequently accessed endpoints.
- Developed and optimized stored procedures in **PostgreSQL** using **PL/pgSQL**, streamlining transactional workflows and improving query execution times for high-volume data operations.
- Implemented secure EHR data exchange standards (HL7/FHIR) to enable seamless, compliant interoperability between healthcare systems, designing scalable data solutions with **Azure Cosmos DB** for low-latency access and high availability.
- Utilized **Entity Framework Core 6** to implement fluent API configurations, database migrations, and **LINQ** query optimizations, reducing **ORM** overhead and enhancing system scalability.
- Applied asynchronous programming patterns using **async/await** to handle long-running operations.
- Designed serverless solutions with **Azure Functions**, integrating with **Event Grid** and **Cosmos DB** to enhance scalability, reduce operational costs, and improve real-time event processing.
- Migrated **Azure Function Apps** from **.NET 6** to **.NET 8**, transitioning from the in-process model to the isolated worker model to improve scalability, maintainability, and alignment with Azure's long-term support strategies.

- Built high-performance **RESTful APIs** with **FastAPI** and **Python**, optimizing real-time data processing for healthcare analytics and integrating **FHIR/HL7** standards for seamless interoperability.
- Developed role-specific, data-visualized modular front-end interfaces with **Angular** and **React** for both administrative and clinical users, leveraging RBAC and real-time data updates to enhance user experience.
- Directed a **front-end** migration from **AngularJS** to **Angular 16**, implementing lazy loading, **NgRx** for state management, and modular components, improving maintainability and load times.
- Designed cross-platform mobile applications for telemedicine using **Xamarin.Forms**, later migrated to **.NET MAUI** to improve performance and maintainability.
- Established **CI/CD** pipelines using **Azure DevOps**, integrating **Terraform** for **Infrastructure as Code (IaC)** and **SonarQube** for code quality analysis, reducing deployment errors and improving software security.
- Implemented **Azure Kubernetes Service (AKS)** and **Docker** for containerized deployments, ensuring high availability and automated scaling.
- Established comprehensive real-time monitoring using **Grafana** and **Azure Monitor**, with advanced logging through **Serilog** and **Azure Application Insights**, reducing downtime and troubleshooting time.
- Integrated several third-party APIs into the healthcare platform using **.NET Core**, enabling secure real-time data exchange with external services and improving system performance and response times.
- Utilized **Swagger/OpenAPI** to generate interactive **API documentation**, streamlining integration efforts for front-end development.
- Designed and implemented a multi-tenant SaaS platform with **ASP.NET Core 6** and **Entity Framework Core**, featuring tenant-based data isolation for security and scalability, reducing infrastructure costs.
- Designed and implemented **GraphQL APIs** using **Node.js** to integrate multiple healthcare data sources, including EHRs and external APIs. Enabled dynamic, real-time data fetching and customized reporting for healthcare professionals by aggregating data from disparate systems into a single, unified API. Optimized scalability and performance using Node.js's asynchronous features for real-time data updates and multi-source querying.
- Drove adoption of **SOLID principles**, **Domain-Driven Design (DDD)**, and **microservices** best practices, improving maintainability and reducing tech debt.
- Led a 5-member **Agile** team on the healthcare platform, managing architecture and cross-functional collaboration, reducing feature delivery time and mentoring junior developers, fostering skill growth, and boosting productivity through pair programming and knowledge-sharing.
- Organized and led knowledge-sharing sessions on best practices for **microservices architecture**, **Azure** deployments, and unit testing, creating comprehensive internal documentation to reduce onboarding times and increase team expertise.
- Improved team efficiency by implementing **agile** development best practices (**Scrum**, **sprint planning**, **Jira**) and fostering mentorship programs.

Emerline, *.NET Full-Stack Developer*

11/2016 – 02/2020

Miami, FL

- Developed and maintained a scalable, cloud-based eCommerce platform using **.NET Core**, **React**, and **AWS**, improving system performance and reliability through optimized API design and efficient database interactions with **Entity Framework Core** and **MS SQL Server**.
- Enhanced real-time data processing by implementing **Apache Kafka** for event-driven architecture, reducing order processing delays and enabling instant inventory updates across distributed services.
- Improved transaction efficiency by optimizing database interactions through **Entity Framework Core** and **T-SQL** stored procedures in **MS SQL Server**. Implemented indexing strategies and partitioning to ensure fast query

execution, reliable data consistency, and high-performance transactional processing.

- Developed and integrated **RESTful APIs** using **Python** and **Django**, facilitating seamless communication between microservices and supporting backend services for real-time analytics, user authentication, and transaction processing.
- Integrated payment gateways like Stripe, PayPal, and Apple Pay using industry-standard authentication mechanisms, including **HMAC** signatures and **PCI-DSS**-compliant tokenization.
- Implemented an advanced search engine using **Elasticsearch** and **ASP.NET Core**, delivering low-latency, high-relevancy product search experiences, increasing user engagement and conversion rates.
- Designed and implemented **AWS cloud services** including **Lambda** for **event-driven workflows**, **S3** for scalable data storage, **SES** for automated email notifications, and **CloudFront** for accelerated content delivery, improving platform efficiency.
- Built a dynamic, responsive, and high-performance frontend using **React**. Implemented state management with **Redux**, optimized performance with lazy loading, code splitting, and server-side rendering (**SSR**), and improved **SEO** and initial load times. Enhanced user experience by reducing page load times and optimizing rendering strategies.
- Automated **CI/CD pipelines** with **Jenkins**, **Docker**, and **Kubernetes**, ensuring seamless deployments, zero-downtime rollouts, and environment consistency across development, staging, and production.
- Implemented role-based authentication and security best practices using **OAuth 2.0**, **AWS Cognito**, and **JWT**, enforcing secure access control and identity management across the platform.

Fjorge, *Full-Stack Developer*

2014 – 10/2016

Minneapolis, MN

- Built **RESTful APIs** using **ASP.NET Web API** to support order processing, inventory updates, and fulfillment workflows. Integrated with third-party eCommerce platforms like Magento and Shopify, enabling seamless data synchronization.
- Managed data across **SQL Server** and **MongoDB**, optimizing query execution using indexing strategies and partitioning for high-performance transactional processing.
- Integrated **RabbitMQ** for event-driven messaging, reducing system latency and ensuring real-time updates across distributed services.
- Built an admin dashboard with **AngularJS** and **Bootstrap**, improving monitoring and analytics for sales, inventory, and fulfillment tracking.
- Developed a customer support portal using **ASP.NET MVC 4** and **C#** to manage customer inquiries, order issues, and return requests. Implemented **Razor** views and **MVC** controllers to provide a server-side rendered, role-based access control (RBAC) system for support agents. Integrated **SQL Server** for customer and order history tracking, allowing efficient issue resolution.
- Built a **WPF**-based desktop application to enhance internal order fulfillment operations, integrating with **ASP.NET Web API** for real-time inventory updates and order tracking. Improved warehouse efficiency by providing a responsive and user-friendly interface tailored for high-volume order processing.
- Deployed applications on **AWS EC2**, used **S3** for file storage, and configured **CloudFront** for optimized content delivery. Contributed to **CI/CD** automation using **TeamCity** and **Git**, ensuring smooth deployments.

Skills

- **Programming Languages:**

C#, JavaScript, TypeScript, Python, Node.js, Java

- **Frameworks & Libraries:**

ASP.NET Core, ASP.NET Web API, ASP.NET MVC, .NET Framework, Blazor, Entity Framework Core, Dapper, WPF, MAUI, FastAPI, Django, Express.js, Nest.js, React (16-18), Next.js, Angular (8, 13, 16), AngularJS, Vue.js, jQuery, MUI, Angular Material, Tailwind CSS, Bootstrap

- **Cloud Platforms & Services:**

Microsoft Azure: Azure Functions, App Service, Blob Storage, SQL Database, Azure Cache for Redis, Azure Cosmos DB, Azure Service Bus, Azure Logic Apps, Azure Event Grid, Azure App Configuration, Azure Kubernetes Service (AKS), Azure Active Directory, Azure Cognitive Services

Amazon Web Services (AWS): Amazon S3, EC2, RDS, Lambda, EKS, CloudWatch, CloudFormation, AWS EventBridge, Amazon SNS, Amazon SQS, DynamoDB, AWS IAM, AWS AppSync, AWS Cognito, AWS CloudFront, AWS CloudTrail

- **Databases & Data Storage:**

SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, Cosmos DB, DynamoDB, Redis, Elasticsearch, Azure Cache for Redis

- **API Technologies:**

RESTful APIs, GraphQL, gRPC, WCF, SOAP, Third-party API integrations, Swagger/OpenAPI, Postman

- **DevOps & CI/CD:**

Jenkins, Azure DevOps, GitHub Actions, GitLab, Docker, Kubernetes, Terraform, AWS EKS

- **Messaging & Event Streaming:**

RabbitMQ, Apache Kafka, Azure Service Bus, Azure SignalR, WebRTC, AWS EventBridge, Amazon SNS, SQS

- **Testing & Quality Assurance:**

xUnit, NUnit, MSTest, Jest, Cypress, Selenium, SonarQube

- **Development Methodologies:**

Test-Driven Development (TDD), Domain-Driven Design (DDD), Behavior-Driven Development (BDD), SOLID, Microservices Architecture, CQRS

- **Version Control & Collaboration Tools:**

Git, SVN, Jira, Confluence

- **Agile Methodologies:**

Scrum, Sprint Planning, Kanban

- **Monitoring & Logging:**

Grafana, Serilog

Education

Bachelor's Degree in Computer Science, *University of North Florida* 2006–2011 | Jacksonville, FL